

1. What is the Protein Data Bank (PDB)?

Answer:

The PDB is a **worldwide repository** of 3D structural data for biological macromolecules (proteins, DNA, RNA, etc.), available freely for download and analysis. It is maintained by **RCSB PDB** and other partners.

2. What kind of information does a PDB file contain?

Answer:

A PDB file contains:

- Atom names and positions (x, y, z coordinates)
- Residue names and numbers
- Chain IDs
- B-factors (temperature)
- Occupancy
- Secondary structure, connectivity, and crystallographic info

3. What is the significance of RCSB PDB?

Answer:

RCSB (Research Collaboratory for Structural Bioinformatics) provides a searchable interface for the PDB, with tools for visualization, sequence alignment, structural comparison, ligand binding, and downloading files.

4. How do you find a protein structure on RCSB?

Answer:

Go to <https://www.rcsb.org>, search using:

- Protein name
- UniProt ID
- PDB ID (e.g., 1A4Y)

- Function or organism

5. What is the format of atom records in a PDB file?

Answer:

Each ATOM line includes:

```
ATOM 1234 CA GLY A 123 11.234 15.678 12.456 1.00 40.00
```

- Atom number, atom name, residue, chain, residue number, x, y, z, occupancy, B-factor.

6. What is macromolecular crystallography?

Answer:

It is a technique where macromolecules are **crystallized** and then bombarded with **X-rays**. The resulting diffraction pattern is used to reconstruct a **3D electron density map** to determine atomic structure.

7. What is a diffraction pattern?

Answer:

A diffraction pattern is the **2D image** produced when X-rays are scattered by a crystal's electrons. It reflects the **positions of atoms** and helps in calculating the 3D electron density.

8. What are B-factors in a PDB file?

Answer:

B-factors (temperature factors) indicate the **atomic displacement or mobility**. Higher B-factors mean more movement or less certainty in atomic position.

9. What is Nuclear Magnetic Resonance (NMR) in structural biology?

Answer:

NMR uses **magnetic fields and radio waves** to detect atomic environments in a molecule, providing distance constraints to determine 3D structure in solution, especially for small proteins.

10. What are the advantages of NMR over crystallography?

Answer:

- No need for crystallization
- Can study proteins in **solution**
- Provides **dynamics** and **flexibility** information
- Suitable for **small to medium proteins**

11. What is Cryo-Electron Microscopy (Cryo-EM)?

Answer:

Cryo-EM is a technique where samples are flash-frozen and observed with **transmission electron microscopy** to visualize macromolecular complexes at near-atomic resolution.

12. When is Cryo-EM preferred?

Answer:

For large complexes, membrane proteins, or flexible structures that are **difficult to crystallize**.

13. What is the resolution of a structure?

Answer:

Resolution defines the level of detail visible in a structure.

- **Low resolution (>4 Å):** general shape
- **Medium (2–4 Å):** side chains and backbone
- **High (<2 Å):** atomic detail

14. How is the molecular weight of a protein estimated from a PDB file?

Answer:

Count all atoms or residues, multiply average amino acid weight (~110 Da), or use PDB tools like **ProtParam** or **PyMOL** plugins.

15. How do you calculate the distance between two atoms in a PDB file?

Answer:

Use the formula:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Can be automated in PyMOL, Chimera, or Python.

16. What does 'HETATM' mean in PDB files?

Answer:

HETATM records represent **non-standard atoms or ligands**, like metal ions, cofactors, or small molecules not part of the protein backbone.

17. How do you visualize a PDB file?

Answer:

Use tools like:

- **PyMOL**
- **Chimera/ChimeraX**
- **Jmol**
- **RCSB 3D View**

18. What is RMSD and how is it related to structure comparison?

Answer:

Root Mean Square Deviation (RMSD) is the average distance between atoms in two superimposed structures. It helps assess **structural similarity**.

19. How can you detect binding sites from PDB data?

Answer:

Look for:

- Conserved residues
- Bound ligands (from HETATM)
- Literature
- Tools like CASTp, SiteMap, or RCSB annotations

20. What is occupancy in PDB files?

Answer:

Occupancy refers to the **fraction of time an atom is observed** in a specific position in the crystal. Values range from 0.00 to 1.00.